Trouble Downstream: How Rivers Carry Rainfall Shocks to Conflict in India

Mahiro Kobayashi [*†‡]

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Abstract

I study how rainfall transmitted through river networks affects violent conflict. For each Indian district, I construct monsoon rainfall over its full upstream drainage area — excluding the district and its neighbours — and route the water downstream with a hydrological model. Transmitted rain raises conflict events by 17 percent per standard deviation the following year; the same monsoon's local rain reduces conflict. I argue the loss travels through the labour market: floods destroy planting, households replace the missing income by supplying more labour just as non-farm earnings contract, and the wage falls — a decline in the return to legal work. The findings imply that local rainfall is the wrong sufficient statistic for climate–conflict studies wherever drainage networks are large.

I. Introduction

A large empirical literature connects rainfall to violent conflict, and nearly all of it measures rainfall in the same place it measures the violence (Miguel et al., 2004; Hsiang et al., 2013; Burke et al., 2015). The implicit assumption is that the economically relevant water is the water that fell overhead. For a country crossed by large river systems that assumption fails on its face: the Ganges at Patna carries the monsoon of Nepal and Uttar Pradesh, not of Patna. If water moves, rainfall shocks are spatially

*Graduate School of Public Policy, University of Tokyo.

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transmitted, and a literature built on local rainfall has measured its treatment with error while missing a channel of transmission entirely.

Published work has moved past the local shocks in three directions, none of which follows the river network. First, a spatial-econometrics literature estimates spillovers across neighbouring grid cells: Harari and La Ferrara (2018) let weather shocks and conflict itself diffuse through distance-based weight matrices. Proximity itself is the channel: the estimates say nearby cells move together, but not what travels between them. Second, a market literature follows prices: world commodity price shocks reach local conflict through what each place produces — coffee and oil in Colombia (Dube and Vargas, 2013), food crops across Africa (McGuirk and Burke, 2020) — so the shock arrives through the price rather than across space. Third, a displacement literature follows people: rainfall scarcity pushes transhumant herders onto neighbouring farmland (McGuirk and Nunn, 2025), and floods displace populations into cities, raising urban disorder (Castells-Quintana et al., 2022). In each strand the shock crosses space, but none of this work links conflict to a weather shock transmitted through the river network.

A smaller literature does look upstream, and defines upstream as the neighbouring district. Duflo and Pande (2007) count dams "upstream of a district" in the adjacent-district sense, and Sarsons (2015), inheriting that definition, shows that downstream of dams, upstream rainfall still moves riots where it no longer moves agricultural income — transmitted water as a threat to the rainfall instrument rather than the object of study. In both designs, what can be seen from upstream is the band of the drainage area bordering the district; in a large basin most of the water comes from further away. Rainfall is also spatially smooth at the district scale — the Indian monsoon especially so — which makes a neighbour's rainfall nearly the district's own: an adjacency measure largely re-reads local rain rather than isolating transmitted water. That bordering band is precisely what this paper's construction excludes, while the network beyond it is the treatment. Closest to the physical carrier, Decet and Marcucci (2025) trace conflict along the river network in Africa, but their hazard is heat: when temperatures rise, conflict moves toward water-rich areas. Here the hazard is the water itself — rainfall collected by the network and delivered downstream — and no existing work measures the water a district actually receives, separates its delivery from its market consequences, or signs local against transmitted rain within one monsoon.

This paper closes the gap in the literature with two contributions. The first is measurement: water rather than rainfall. For each Indian district, monsoon rainfall is aggregated over the upstream catchment traced through 21,622 sub-basins (Lehner and Grill, 2013), each weighted by how far its water travels along the river to reach

the district, excluding every basin that overlaps the district or its neighbours, so the treatment is rain that fell strictly elsewhere; and the rain is then followed downstream with CaMa-Flood (Yamazaki et al., 2011), a river-routing model brought in from hydrology, which routes the catchment's runoff — rainfall after the land surface has taken its share — into the discharge arriving at the district's own rivers: the downstream hazard itself, not a rainfall proxy for it. The second contribution signs local against transmitted rain in one regression: lagged upstream monsoon rainfall raises conflict intensity with an elasticity of 0.91 — a one-standard-deviation shock raises expected UCDP event counts 17 percent the following year — while rain on the district itself enters negatively.

The results provide suggestive evidence for the mechanism. Upstream rain arrives at the district as flood, carried by the river, and first destroys the kharif crop — planting at the margin, not the yield of surviving fields. For households that cannot insure the loss, destruction is a labour-supply shock: they replace the missing income by selling more labour to the casual market (Kochar, 1999; Rose, 2001). The missing income need not be the crop alone, either: casual non-farm earnings fall at the same horizon. Employment rises while the wage falls — a supply shift, not prosperity, and the equilibrium wage response that poor, credit-constrained labour markets amplify (Jayachandran, 2006) — and the return to legal work drops on both margins at once. The results also show that conflict responds to upstream monsoon rainfall only from the following January — months after the lost harvest would have been sold.

Two limitations are stated up front. First, the downstream side of the design is unresolved. Rainfall on the catchments downstream of a district enters negatively rather than at zero, and the estimates cannot yet separate the two readings — a coefficient that partly picks up rainfall correlated with the district's own, or a genuine effect of downstream water; if the effect is real, its mechanism is an open question. Second, the roles of infrastructure and policy have not been explored extensively. Whether dams could blunt the shock is not yet answerable — they sit on the largest rivers, so the observational contrast cannot separate what a dam does from where it is — and the employment guarantee that moderates local rainfall's conflict effect (Fetzer, 2020) does not reach damage that local relief triggers never see; Section VII reports where that evidence stands.

Two literatures locate the contribution. The climate–conflict literature finds local weather moving violence at every scale (Miguel et al., 2004; Hsiang et al., 2013; Harari and La Ferrara, 2018), in India included (Vanden Eynde, 2018; Sekhri and Storeygard, 2014) and Indonesia (Gatti et al., 2021), though the aggregate reading remains contested (Couttenier and Soubeyran, 2013; Mach et al., 2019; Burke et al., 2015). To it this paper adds the transmitted component: one monsoon enters twice,

and its transmitted part moves violence in the opposite direction to its local part, so a design built on local rain alone estimates a blend of channels that partly cancel. The opportunity-cost literature ties fighting to income (Dal Bó and Dal Bó, 2011; Dube and Vargas, 2013; McGuirk and Burke, 2020); here the fall in the return to legal work is observed rather than inferred from output: the wage falls while employment rises — the joint movement that marks a labour-supply response to a destroyed harvest — and household surveys show the squeeze on the farm and non-farm margins at once.

Section II describes the data; Section III the construction and identification; Section IV the main result and its inference battery; Sections V–VII the flood channel, the mechanism, and infrastructure and policy; Section VIII concludes. Online Appendix C documents the routed-water construction.

II. Data and Descriptive Results

A. Violent Conflict

The outcome is the district-year count of violent events from the UCDP Georeferenced Event Dataset (Sundberg and Melander, 2013), version 26.1, geocoded to 2001-census district boundaries. UCDP defines an event as the use of armed force by an organized actor against another organized actor or against civilians, causing at least one direct death at a specific location and date; events enter only if the actors belong to a conflict that has crossed twenty-five deaths in a calendar year, in one of three categories: state-based conflict, non-state conflict, and one-sided violence against civilians.[1] Three consequences of this definition matter here. First, the one-death floor means the count measures lethal incidents, not unrest generally. Second, the death must be direct, so mortality from the floods the treatment delivers — drowning, displacement, disease — does not enter the outcome. Third, the actors must be organized, so communal violence enters only as lethal two-sided clashes between communal groups (the communal category of Section IV). The outcome is therefore organized armed violence: insurgency (the state's fighting with armed groups and those groups' attacks on civilians) accounts for 91 percent of events, and clashes between communal groups for 4 percent. Events located only to a state or coarser (10.3 percent of the 1990–2025 record) are not included in the analysis.

[1] State-based conflict: armed force between two parties, at least one the government of a state, over a contested incompatibility concerning government or territory, causing at least twenty-five battle-related deaths in a calendar year. Non-state conflict: armed force between two organized armed groups, neither the government of a state, causing at least twenty-five battle-related deaths in a year. One-sided violence: armed force by a government or formally organized group against civilians, causing at least twenty-five civilian deaths in a year; extrajudicial killings in custody are excluded. In the estimation sample, 65.0 percent of events are state-based, 4.7 percent non-state, and 30.3 percent one-sided.

B. *Rainfall, Temperature, and Hydrography*

Rainfall.—Precipitation comes from CHIRPS version 3 (Funk et al., 2026), a 0.05-degree daily product that blends satellite estimates with station observations. It suits the design better than gauge-only grids, reanalysis, or interpolated climatologies: its satellite component reaches the upstream catchments in Nepal, China, Bhutan, and Pakistan that India's gauge-based grid does not cover, its resolution matches the sub-basins from which catchments are built, and it spans the whole sample. Online Appendix A.4 sets out the choice and re-estimates the headline under five alternative products.

Temperature.—The area-weighted district mean over the same window comes from CHIRTS-ERA5 (Funk et al., 2019; Verdin et al., 2020), which merges the Climate Hazards Center's station-calibrated temperature record with ERA5 daily variability on the same 0.05-degree grid as CHIRPS.

Hydrography.—The drainage network is HydroBASINS level 10 (Lehner and Grill, 2013): 21,622 sub-basins covering the Indian subcontinent and its transboundary catchments, each linked to the sub-basin it drains into. I use it to trace each district's upstream catchment and to apply the neighbour exclusion of Section III.A, which drops every sub-basin overlapping the district or a neighbouring district so that the treatment does not mix in local weather, and its coarser levels to cluster standard errors by river system (Section IV); Online Appendix A.2 explains the choice of level.

C. *Dams and Routed Water*

Dams.—Data on dams come from the National Register of Large Dams, the Central Water Commission's census of India's 5,751 large dams, where "large" follows the register's criteria: at least 15 meters high, or 10–15 meters with additional conditions on crest length, reservoir capacity, or spillway discharge. The register reports each dam's location in degrees, minutes and seconds, which are parsed directly rather than matched to an external reservoir database; 5,736 entries (99.7 percent) carry a readable coordinate, and every one falls on the HydroBASINS network, so 3,637 of the 3,638 dams commissioned by 1989 (227 billion cubic metres of gross storage) are placed on the drainage network.

Routed water.—Rainfall on an upstream catchment is a proxy for the water that actually arrives downstream, so the analysis also uses a physically routed measure: daily discharge simulated over 1989–2025 by the CaMa-Flood river-routing model (Yamazaki et al., 2011). The model moves runoff from ERA5-Land, a land-surface simulation driven by the weather alone, along a fixed river and floodplain map built from MERIT Hydro. It does not represent dams, reservoirs, barrages, canals, or levees, so its only input that changes over time is runoff: once district and year effects

are absorbed, the variation in routed water comes from weather, not from how the water is managed. Runoff within 50 kilometres of each district is removed before routing — the 50-kilometre mask, the routing counterpart of the neighbour exclusion — so the routed water is upstream-origin by construction, and the series is summarised as *flood-stage days*, days on which discharge exceeds its local two-year return level. Online Appendix C documents construction and validation in detail.

D. Sample Construction and Summary Statistics

The main estimation sample is fixed by two restrictions. First, of the 593 districts in the 2001-census boundary file, 168 have no upstream catchment left after the neighbour exclusion — both defined in Section III.A — and so receive no treatment. Second, of the remaining 425, 215 record no conflict event between 1990 and 2025. Since the fixed-effects Poisson, the estimation method used in this paper, conditions on each district's total count, these all-zero districts carry no information about the coefficient and drop out by construction. The headline estimates therefore use 210 districts and 7,560 district-years.

Table 1 reports the summary statistics of the headline variables. With district fixed effects, the identifying variation is a standard deviation of about 172 mm around each district's own mean, against 351 mm in the raw cross-section. Panel B separates the monsoon from the rest of the year: upstream catchments receive 1,002 mm between June and September against 290 mm in the other eight months, and the monsoon carries almost twice the within-district variation (172 mm against 89 mm). Online Appendix Figure E.4 shows that 78 percent of upstream rainfall falls in the monsoon, with nearly the same monthly profile as the district's own rainfall. A district-year experiences 1.001 events on average, with a median of zero and a maximum of 197; roughly four in five district-years record none (incidence 0.17). Unlike rainfall, conflict is not concentrated in the monsoon: only 36 percent of events fall between June and September. Panel C's routed-water measures cover fewer district-years (6,336).[2] They are far more concentrated than rainfall, too: in an average year a district's river exceeds its two-year level — the flow its annual peak reaches about one year in two — on 2.3 days and its five-year level on 0.6, and in 79 percent of district-years it never reaches the two-year level.

Figure 1 maps the three objects. Conflict is spatially concentrated — a Maoist belt through central and eastern India, the Kashmir valley, and the Northeast —

[2]Routed water is defined for 176 of the 210 estimation districts. Of the rest, 15 contain no river channel the routing model resolves, 3 are headwater districts left with almost no upstream input once the 50-kilometre mask removes their nearby runoff, and 16 lie below a river split, where removing nearby runoff shifts how the flow divides between branches, so the routed measure is not reliable there.

Table 1: Summary statistics, estimation sample

	N	Mean	SD	Within-district SD	Min	Max
A. Conflict						
Conflict events	7,560	1.001	5.638	1.729	0.000	197.000
Conflict incidence (0/1)	7,560	0.173	0.378	0.299	0.000	1.000
B. Rainfall and temperature						
Monsoon (June–September)						
Upstream rainfall (mm)	7,560	1,001.776	350.989	172.122	62.793	3,182.764
Local rainfall (mm)	7,560	1,054.856	512.680	200.528	21.175	4,323.204
Upstream temperature (°C)	7,560	29.328	4.536	0.552	13.143	36.962
Local temperature (°C)	7,560	32.862	2.711	0.569	19.126	39.017
January–May and October–December						
Upstream rainfall (mm)	7,560	289.816	211.082	88.948	6.763	1,604.946
Local rainfall (mm)	7,560	334.024	317.905	109.456	1.476	2,208.725
Upstream temperature (°C)	7,560	26.043	6.987	0.543	3.247	34.902
Local temperature (°C)	7,560	29.937	4.084	0.552	4.119	36.404
C. Routed water (CaMa-Flood)						
Upstream flood-stage days, 2-year level	6,336	2.324	7.209	3.818	0.000	75.000
Upstream flood-stage days, 5-year level	6,336	0.574	2.945	1.502	0.000	44.000
Peak discharge / 2-year level	6,336	0.744	0.489	0.306	0.000	9.700

Notes: District-year observations on the Poisson estimation sample — the 210 districts recording at least one located conflict event over 1990–2025. Conflict counts events UCDP places to a district or better; events located only to a state are excluded throughout. Panel B's second block is the calendar year less the monsoon, so it is not a contiguous season: it runs January–May and October–December of the same year. 'Within-district SD' is the mean across districts of the within-district standard deviation, and is the variation the fixed-effects design uses. Upstream rainfall is the river-distance-weighted total over the catchment after the neighbour exclusion (Section III). Panel C is routed through CaMa-Flood and is available for 1990–2025 on the districts carrying a river cell and an upstream catchment, so it covers fewer district-years than Panels A and B; a flood-stage day is a day whose centred three-day mean discharge exceeds the cell's fitted return level.

while upstream rainfall is high along the Himalayan front and the western coast and low across the northwest. Panel C shows the districts in the estimation sample.

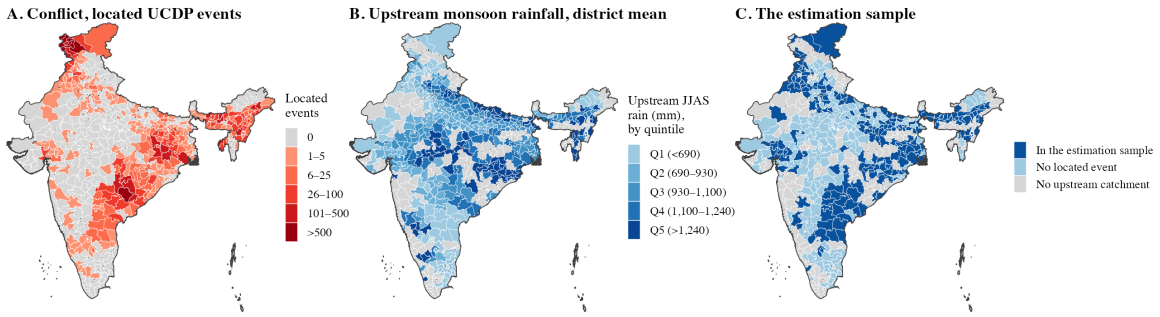


Figure 1: Conflict, upstream rainfall, and the estimation sample

Notes: Panels A and B describe all 593 mapped districts, not only those estimated on. Panel A shades each district by its located UCDP events over 1990–2025; Panel B by its June–September upstream catchment rainfall over the same years, in quintiles of the district means, with the legend's bounds rounded to the nearest ten millimetres. Panel C marks the 210 districts that enter the headline regression, carrying 7,564 located events between them, and separates the two reasons the rest are out: 168 have no upstream catchment and so no treatment, while the remainder have one but record no located event in the window and are dropped by the Poisson's district fixed effects, which cannot estimate on a district whose outcome is zero throughout. Grey marks a count of zero in Panel A and a district with no upstream catchment in Panels B and C. Treatment is upstream rainfall, so Panel B is not a map of where it rains but of what falls on the catchments draining into each district.

III. Empirical Strategy

A. *Constructing Rainfall from Elsewhere*

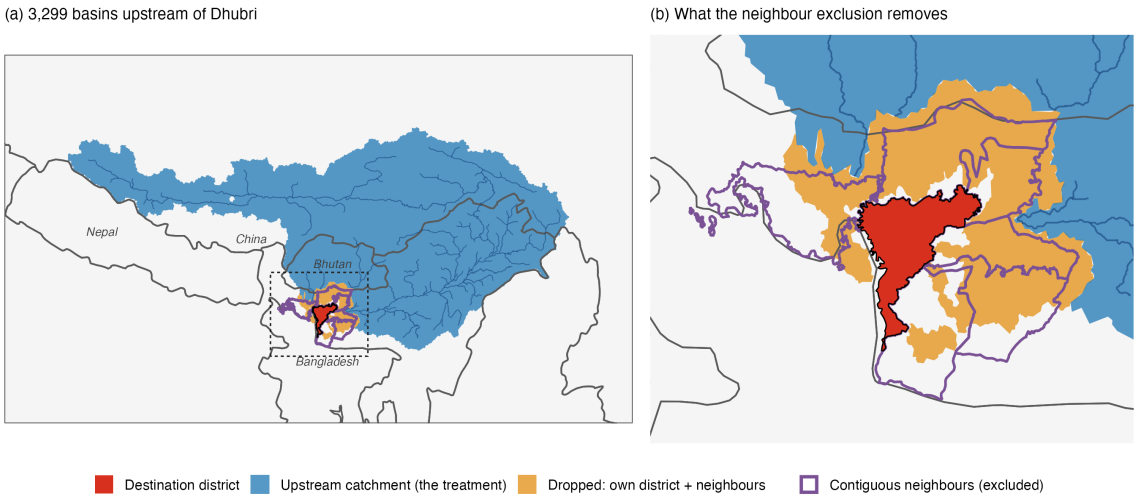


Figure 2: The upstream treatment, for one district

Notes: Panel (a): the 3,299 sub-basins that drain into Dhubri, Assam, the district at the 90th percentile of the estimation sample by number of upstream sub-basins — a catchment reaching across Bhutan and into Tibet, hundreds of kilometres from the district itself. Panel (b) enlarges the dashed box. The treatment is monsoon rainfall over the blue area only, each basin weighted by the inverse of its distance to the district along the river: basins overlapping the district, any of its 6 contiguous neighbours (purple outline) or the adjacent foreign administrative units are removed, which drops 111 basins (amber). Identifying variation is therefore rain that fell neither on the district nor on anywhere it touches.

For district i, let $U(i)$ denote the set of sub-basins[3] that drain into i through the hydrological network, *excluding* every sub-basin that overlaps i itself or any district sharing a border with i. Across an international border, the adjacent administrative unit of the neighbouring country counts as a bordering district. This neighbour exclusion is the preferred choice, and Figure 2 visualizes what it does: it removes the district's own weather and its immediate neighbourhood's weather from the treatment, so that identifying variation comes from rain the district never saw. Upstream rainfall is a weighted average in which each sub-basin counts by how far its water must travel along the river:

$$R_{it}^{UP} = \sum_{b \in U(i)} w_{ib}\, R_{bt}, \qquad w_{ib} = \frac{\omega_{ib}}{\sum_{b' \in U(i)} \omega_{ib'}}, \qquad \omega_{ib} = \sum_{p \in P(i,b)} \frac{s_p}{1 + r_p},$$

where R_{bt} is sub-basin b's monsoon rainfall, $P(i,b)$ is the set of river paths from sub-basin b to district i, r_p is the length in kilometres of path p along the network, from sub-basin b's outlet to where its water enters district i, and s_p is the share of b's flow that follows p (Online Appendix Figure A.4 draws each piece of this notation). On a

[3]HydroBASINS level 10 throughout; Online Appendix A.2 explains the choice of level.

single channel there is one path and $s_p = 1$, so $\omega_{ib} = (1 + r_{ib})^{-1}$; where a river splits, the branch carries the share of the flow it receives in a hydrological routing model (Online Appendix A.3). The weight therefore decays with distance along the river, and Online Appendix A.3 shows that the estimates are not sensitive to this choice.

B. Specification

The estimating equation is a fixed-effects Poisson regression,

$$\mathbb{E}[A_{it} \mid \cdot] = \exp\Big(\beta_1 \log R^{UP}_{it} + \beta_2 \log R^{UP}_{i,t-1} + \mathbf{u}'_{it}\boldsymbol{\gamma} + \mathbf{x}'_{it}\boldsymbol{\theta} + \rho\, \bar{A}_{i,t-1} + \bar{\mathbf{x}}'_{it}\boldsymbol{\lambda} + \alpha_i + \delta_t\Big), \quad (1)$$

where A_{it} is the district-year count of UCDP events, $\mathbf{u}_{it}$ is current and lagged upstream temperature, $\mathbf{x}_{it}$ is current and lagged local rainfall and temperature, and α_i, δ_t are district and year effects. A bar denotes a neighbourhood average: for any district-level variable v, $\bar{v}_{it} = \sum_{j \in N(i)} W_{ij} v_{jt}$, where $N(i)$ is the set of districts sharing a border with i and the weights W_{ij} are row-normalised so that they sum to one.[4] Thus $\bar{A}_{i,t-1}$ is neighbours' lagged conflict, entered as $\log(1 + \text{count})$, and $\bar{\mathbf{x}}_{it}$ is neighbours' local weather.

Equation (1) is a variant of the spatial Durbin model that Harari and La Ferrara (2018) use for conflict: besides the district's own regressors, it includes its neighbours' conflict and weather, so that violence spreading across district borders and weather shocks common to a region are not attributed to upstream water. Equation (1) omits three terms of the preferred model of Harari and La Ferrara (2018). The district's own conflict in the previous year and neighbours' conflict in the same year cannot be estimated consistently in this setting: the first is entangled with the district fixed effect in a fixed-effects Poisson model, and the second is determined jointly with the outcome. Neighbours' conflict therefore enters with a one-year lag, $\bar{A}_{i,t-1}$. Neighbours' upstream rainfall is left out since neighbouring districts drain overlapping catchments, which makes it nearly a copy of the district's own (correlation 0.91 after fixed effects). Online Appendix B.3 adds the three terms back, one at a time and all together, to check that these omissions do not drive the result.

Local weather is essential here to isolate the effect of upstream rainfall. Upstream and local monsoon rainfall correlate at 0.71 after fixed effects, because the same weather system waters a district and its catchment. They act through different channels: local rain directly, on the district's own crops and roads; upstream rain plausibly through the river. Holding local weather fixed isolates the transmitted channel

[4] So $\bar{v}_{it}$ is simply the average of v across district i's immediate neighbours in year t, and district i itself is never among them: $W_{ii} = 0$. Row-normalising means a district with three neighbours and one with eight are each compared with the average of their own, rather than the district with more neighbours receiving a mechanically larger term.

through the river.

C. *Identification and Inference*

Upstream rainfall is a shift–share,

$$R_{it}^{UP} = \sum_{b \in U(i)} w_{ib}\, R_{bt}, \qquad \sum_{b} w_{ib} = 1,$$

with exposure shares w_{ib} fixed by hydrography and shocks R_{bt} the basin-year rainfall. Identification rests on the exogeneity of the shocks, not the shares (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2022). Let μ_i be district i's expected exposure, the treatment it would receive on average over draws of basin rainfall (Borusyak and Hull, 2023). The identifying assumption is

$$\mathbb{E}\big[(\log R_{it}^{UP} - \mu_i)\, \varepsilon_{it}\big] = 0 \tag{2}$$

after district and year effects and local weather are partialled out: whether a district's catchment receives more or less rain than usual in a given year is unrelated to the other, unobserved causes of conflict in that district that year. The treatment is built from basin rainfall, so this is in substance a condition on the shocks: rainfall above or below a basin's climatology must be unrelated to the unobservables of the districts it drains (Borusyak et al., 2022). It is an assumption about weather, not about where rivers run, and it identifies from deviations from climatology in the manner standard to the climate–economy literature (Dell et al., 2014).

Two things follow. First, expected exposure must not bias the estimate. When climatology is stable, μ_i is constant and district effects absorb it; the demanding test lets climatology drift and recenters on a decade-specific μ_i (Borusyak and Hull, 2023). This is also the margin on which Christian and Barrett (2024) caution that weather–conflict panels can manufacture correlations out of slow-moving common components. Second, the treatment's variation comes from basin rainfall, so the estimate is consistent as the number of independent basin shocks grows, not the number of districts (Borusyak et al., 2022). Basin rainfall is spatially correlated within a season, so the effective number of independent shocks — the inverse Herfindahl of basin weights — is 37.4 across the 82 major sub-catchments and 7.20 across the 21 macro basins (Online Appendix B). Treating districts as independent would overstate precision. Standard errors therefore cluster on whole river systems and treat basin shocks as the sampling unit (Adão et al., 2019), and randomization tests redraw the shocks themselves. Section IV and Online Appendices B.4 and B.5 report both.

IV. Upstream Rainfall and Violent Conflict

This section reports the paper's central finding. Rain that falls on the catchments upstream of a district raises violent conflict in that district the following year, while the same monsoon's rain falling on the district itself lowers it — two opposite responses to one weather system, separated only by where the water lands. The effect is concentrated in insurgent violence against civilians rather than in battles between armed forces, is carried by the monsoon months rather than by winter rainfall, and survives every inference scheme and randomization test I impose on it.

Table 2 reports the main results. Column 1 omits the spatial controls and is the conservative estimate. Upstream monsoon rainfall raises the expected number of conflict events next year with an elasticity of 0.91 (Conley standard error 0.36), while contemporaneous upstream rainfall is small and insignificant. Local rainfall in the same regression enters negatively: rain measured upstream and rain measured where the violence occurs move conflict in opposite directions, implying that regressions using local rainfall alone conflate two channels. A 10 percent wetter upstream monsoon raises expected events the following year by about 9.1 percent; one standard deviation of the identifying variation — a monsoon 19 percent wetter — raises them by about 17 percent, against a mean of 1.00 events per district-year[5].

Columns 2–4 add spatial controls, which sharpen the estimate. Holding neighbours' lagged conflict fixed moves the elasticity to 0.99; holding neighbours' own weather fixed moves it to 1.35. Neighbours' weather belongs in the specification because the monsoon is spatially smooth: a district's upstream exposure still correlates 0.75 with its neighbours' local rainfall after every basin touching them is excluded, and neighbours' rain carries the pacifying local-income sign, so omitting it biases the estimate toward zero. That smoothness — rainfall changing little from one district to the next — also makes local and neighbours' rainfall nearly collinear once both enter, so from column 3 onward local rainfall serves only as a control and its coefficient is not separately interpretable. With both sets of controls the elasticity is 1.46 (Conley 0.28). The elasticities should not be compared across columns as they stand, since each spatial control partials variation out of the treatment itself. The two rows above the sample counts do the rescaling: each column is translated into the effect of a one-standard-deviation wetter monsoon using the variation that column identifies from, and on that common footing the rising coefficient is not a rising effect.

[5]The two readings are one coefficient at two scales. A Poisson coefficient on log rainfall is an elasticity, so for small changes the percentage response is 0.91 times the percentage change in rainfall; scaling to a standard deviation instead asks how much the shock matters given the variation that exists, and $\exp(0.91 \times 0.17) - 1$ makes that conversion exact. Table 2 reports both for every column, which matters because each spatial control partials variation out of the treatment and so moves the standard deviation without moving the elasticity.

Table 2: Upstream rainfall and conflict

	Baseline	Spatial controls			
	(1)	(2)	(3)	(4)	(5)
log Upstream Precip. (t)	-0.045	-0.106	-0.047	-0.086	-0.275
	(0.371)	(0.412)	(0.381)	(0.487)	(0.468)
log Upstream Precip. $(t-1)$	0.910**	0.994***	1.351***	1.460***	1.242***
	(0.359)	(0.314)	(0.351)	(0.283)	(0.277)
log Local Precip. (t)	-0.353	-0.385	-0.103	-0.088	-0.675
	(0.258)	(0.305)	(0.482)	(0.579)	(0.658)
log Local Precip. $(t-1)$	-0.921***	-0.978***	0.527	0.507	0.567
	(0.324)	(0.330)	(0.921)	(1.025)	(1.012)
log Neighbours' Conflict $(t-1)$		1.002***		0.994***	0.957***
		(0.149)		(0.149)	(0.155)
log Neighbours' Local Precip. (t)			-0.188	-0.315	1.701*
			(0.615)	(0.871)	(1.009)
log Neighbours' Local Precip. $(t-1)$			-2.046*	-2.103	-1.443
			(1.190)	(1.316)	(1.412)
log Downstream Precip. (t)					-1.329***
					(0.456)
log Downstream Precip. $(t-1)$					-0.906***
					(0.336)
1 SD of identifying variation (log points)	0.17	0.17	0.12	0.12	0.11
equivalently, a wetter monsoon of (%)	19	19	12	12	12
Implied effect of a 1 SD wetter monsoon (%)	17	19	17	18	15
Observations	7,560	7,560	7,560	7,560	6,372
Number of Districts	210	210	210	210	177
Mean of dependent variable	1.00	1.00	1.00	1.00	1.06
Estimation	Poisson	Poisson	Poisson	Poisson	Poisson
District & Year FE	✓	✓	✓	✓	✓

Notes: The outcome is the count of UCDP events located to a district or better, estimated by Poisson on 1990–2025, the longest frame the data allow. Temperature controls — upstream, local, neighbours' and downstream, current and lagged — enter their respective columns. Column 2 adds neighbours' lagged conflict; column 3 adds neighbours' current and lagged local rainfall and temperature; column 4 adds both. Neighbours' variables are averages over the districts sharing a border under row-normalised queen contiguity, conflict entering as $\log(1 + \text{count})$. Column 5 adds the downstream weather, built by the same construction following the flow network in the opposite direction, and is estimated only on districts where that catchment is defined; on that sample the coefficient estimate for log Upstream Precip. (t-1) in the column 4 gives 1.36. The contemporaneous spatial lag and the own lagged conflict term are omitted throughout; Online Appendix B.3 gives the reasons. Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Column 5 adds downstream rainfall, built identically but following the flow network in the opposite direction. Its coefficients are negative, and the upstream estimate is unchanged on that sample. The downstream term is not the clean zero a placebo would ideally return; Online Appendix E examines it and Section VIII discusses what it means.

Types of Conflicts.—The outcome in Table 2 pools several types of violence and several conflict systems. Online Appendix E.2 cuts it each way on its own: no single type carries the result, and by system it rests on the Northeast and communal columns, with Maoist violence — the largest component of the sample, and the canonical setting for an agricultural-labour account of recruitment (Shah, 2018; Khanna and Zimmermann, 2017) — indistinguishable from zero, against Vanden Eynde (2018), who find

Table 3: Which conflicts respond: conflict system by type of violence

	Maoist		Kashmir		Northeast	
	State-based	One-sided	State-based	One-sided	State-based	One-sided
	(1)	(2)	(3)	(4)	(5)	(6)
log Upstream Precip. $(t-1)$	0.260	2.593***	0.281	0.524	1.230***	1.560
	(0.672)	(0.941)	(0.209)	(0.816)	(0.475)	(1.590)
log Upstream Precip. (t)	-0.954*	-1.130	-0.739	0.286	1.064	-1.515
	(0.534)	(1.181)	(0.814)	(0.292)	(1.185)	(1.212)
log Local Precip. (t)	-0.516	-2.019	0.441	-2.179***	-0.052	0.998**
	(0.743)	(1.553)	(0.391)	(0.648)	(0.634)	(0.453)
log Local Precip. $(t-1)$	-0.238	-3.005**	0.841	1.724	-0.328	-1.790
	(1.064)	(1.226)	(0.804)	(1.237)	(0.244)	(1.117)
1 SD of identifying variation (log points)	0.10	0.10	0.13	0.14	0.09	0.09
equivalently, a wetter monsoon of (%)	11	10	14	15	9	9
Implied effect of a 1 SD wetter monsoon (%)	3	28	4	7	12	14
Observations	2,835	2,070	1,044	1,258	1,190	644
Districts	81	69	29	37	34	23
Mean of dependent variable	0.66	0.60	1.52	0.42	1.02	0.68
Estimation	Poisson	Poisson	Poisson	Poisson	Poisson	Poisson
District & Year FE	✓	✓	✓	✓	✓	✓

Notes: Each column is a separate Poisson regression on 1990–2025 under the specification of Table 2, column 4: district and year fixed effects, the full weather controls, and the spatial block. Only the outcome changes — events of one conflict system and one UCDP type of violence. Neighbours' conflict in the spatial block is built from the column's own outcome. Communal violence is non-state by definition and so has no column here. Poisson drops districts recording no events of a given kind, which is why the district counts differ across columns; the columns are therefore not a decomposition of any single total. Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Naxalite violence responds to local rainfall. Table 3 crosses the two and narrows that null: it holds for the pooled Maoist count only, since one-sided violence against civilians responds at 2.59 (standard error 0.94) while state-based violence does not (0.26, 0.67). The Northeast responds in both, again more in one-sided violence (1.56 against 1.23), and Kashmir little in either. An opportunity-cost account of recruitment predicts more fighting when farm income falls, not a response confined to violence against civilians; the pattern fits a damaged agricultural economy leaving civilians more exposed to predation, and matches the compositional shift inside the flood window in Section VI. For communal violence, Mitra and Ray (2014) find riots respond to relative rather than absolute prosperity, so an unevenly falling shock can raise it where opportunity-cost logic predicts less. Section VI takes up what the agricultural data can test.

The seasonal window.—The monsoon — June to September — is the window because it delivers most of the year's rainfall and drives the flooding the mechanism runs through; the agricultural calendar is secondary. Those months are also when the kharif crop is in the ground, but the window is chosen for the water rather than for the crop, and the two do not coincide exactly: the monsoon is a fixed calendar window while the kharif season starts and ends at different dates across districts and crops. Table E.2 asks whether that choice discards anything, using MIRCA2000 monthly cropped-area shares to weight rainfall by when each district's own crops are in the ground.

Weighting the whole year that way gives a positive and significant but smaller elasticity (0.75); the two regressors correlate 0.89 within district, and the added months mostly dilute the monsoon signal. The sharper test isolates what the monsoon window leaves out. The rabi portion of the calendar — October to May, when the winter crop is grown — correlates only 0.07 with monsoon rainfall, enters at -0.066, and the growing-season component orthogonal to the monsoon at -0.32; in both columns the monsoon coefficient is unchanged. Rainfall outside the monsoon, even weighted by each district's own crop calendar, has no detectable effect: the response is a monsoon response.[6]

The buffer radius.—The neighbour exclusion keeps local and neighbouring districts' rainfall out of the upstream measure, but the boundary it draws is one point on a continuum. Online Appendix Figure A.3 sweeps that continuum with a buffer radius instead: at each radius from 0 to 100 kilometres, every sub-basin within that distance of the district is dropped, the upstream rainfall exposure and the upstream temperature control are rebuilt and the specifications of Table 2, columns 1 and 4, are re-estimated, so that only the catchment boundary varies. The neighbour exclusion is topological — it drops basins that touch the district or any district bordering it — so no buffer radius reproduces Table 2. The figure therefore asks whether the result depends on where a distance boundary is drawn, not whether it can be recovered at one. Under the spatial specification, the lagged upstream elasticity is positive at every cell (Online Appendix Figure A.3), ranging from 0.52 to 2.28, and 37 of the 44 cells clear the Conley 95 percent interval. Without the spatial block, the elasticity is smaller and, at radii below about 80 kilometres, imprecise. This is the omitted-variable pattern of Table 2: nearby rainfall carries the pacifying sign and pulls the estimate toward zero when it is left out. Widening the buffer radius also removes districts whose catchment empties — from 296 to 176 under polygon distance — so the lower row of the figure holds the district set fixed at the survivors of the 100-kilometre cutoff. On that fixed set, the elasticity declines as the buffer radius widens, from 2.28 to 0.79 under spatial controls; Online Appendix Figure A.3 plots the whole sweep, and Online Appendix A.2 reads the two ends of it as bounds. Nearer basins thus contribute more of the response, as they should if the shock is transmitted by water and attenuates with distance. Online Appendix A.2 shows what each buffer radius removes and re-estimates the headline under alternative weightings, straight-line distance and runoff volume among them: pushed inward, the treatment absorbs the district's own weather; pushed outward, the district–catchment map loses the variation that identifies the effect.

[6]Counting conflict over the monsoon year (June t to May $t+1$) rather than the calendar year moves the response from the lagged to the contemporaneous term (0.91 to 0.43 and -0.03 to 0.53 on 1990–2024, ending with the last monsoon year observed in full) without changing its size; the calendar-year count is kept so that Tables 2, 3, and E.2 share one outcome.

Inference.—Districts along the same river share upstream basins, so their treatments are mechanically correlated and district-clustered standard errors may be too small (Adão et al., 2019). Online Appendix B.4 re-prices the estimate under progressively coarser dependence structures — spatial and network kernels, exposure-robust standard errors, and clustering on whole river systems — and adds four randomization tests, which permute the district–catchment map and the timing of the shocks in turn (Online Appendix B.5 describes each). The estimate clears one percent under every variance estimate — the least favourable still puts it more than three standard errors from zero — and five percent under every randomization test, the weakest giving $p = 0.0098$ (Online Appendix Figure E.8 reports all four).

V. Routed Water: The Flood Channel

An upstream rainfall shock first reaches a district as water in the river. Routing runoff for 1989–2025 through the CaMa-Flood river model (Yamazaki et al., 2011), with the 50-kilometre mask removing the district's own and nearby runoff so the routed water is upstream-origin by construction, converts the rainfall shock into the physical quantity the district actually receives: a log point of upstream monsoon rainfall adds 4.8 days above the local two-year return level (standard error 0.47).

Table 4 races the rainfall total against the routed flood-stage days on one sample. Contemporaneously the rainfall total does nothing while flood-stage days carry 0.020; inundated area and flood depth carry nothing beside them (Online Appendix C). Within the year the shock arrives as a hazard, and the margin that matters is the river standing high, not the volume delivered. One qualification: Section VII shows that water dams held back — water that never arrived — predicts conflict about as strongly as water that did, so flood-stage days track how severe the flood season was upstream as well as what was delivered.

What the water does while it is on the ground is also distinct from what follows it. Dating flood-stage months exactly with the daily routed series, one-sided violence against civilians rises during the flood month (0.020 per flood-stage day) while state–insurgent battles do not move (0.004, standard error 0.011), and the aggregate rises in the month after (0.019): what increases while the water stands is violence against disrupted populations. The response is also larger, though not significantly, where the physical account says it should be — in districts whose river cells run flat, so that arriving water spreads rather than passes (0.28 per standard deviation of flat-cell share, standard error 0.19) — the direction the mechanism predicts.

The lagged response is a different animal. No flood measure absorbs it: the lagged rainfall elasticity is 0.93 alone and 0.87 with routed days included, a movement of 6 percent (1.50 and 1.47 with the spatial block). Conditioning on the physical water that

Table 4: What arrives: upstream rainfall against routed flood-stage days, one sample

	Rainfall		Routed water		Both	
	(1)	(2)	(3)	(4)	(5)	(6)
log Upstream Precip. (t)	-0.017	-0.045			-0.083	-0.112
	(0.398)	(0.500)			(0.411)	(0.507)
log Upstream Precip. ($t-1$)	0.933**	1.505***			0.874**	1.469***
	(0.395)	(0.311)			(0.396)	(0.307)
Upstream Flood-Stage Days (t)			0.020**	0.018**	0.020**	0.018**
			(0.008)	(0.008)	(0.009)	(0.008)
Upstream Flood-Stage Days ($t-1$)			0.028***	0.015**	0.025**	0.011
			(0.010)	(0.007)	(0.011)	(0.007)
log Local Precip. (t)	-0.356	-0.055	-0.390	-0.121	-0.336	-0.044
	(0.278)	(0.614)	(0.295)	(0.610)	(0.287)	(0.617)
log Local Precip. ($t-1$)	-0.893***	0.483	-0.260	-0.000	-0.894**	0.457
	(0.346)	(1.099)	(0.225)	(1.063)	(0.351)	(1.104)
1 SD of identifying variation, rainfall (log points)	0.12	0.11	—	—	0.11	0.11
Implied effect of a 1 SD wetter monsoon (%)	11	18	—	—	11	17
1 SD of identifying variation, flood-stage days	—	—	5.78	5.77	5.74	5.73
Implied effect of a 1 SD rise in flood days (%)	—	—	12	11	12	11
Observations	6,336	6,336	6,336	6,336	6,336	6,336
Number of Districts	176	176	176	176	176	176
Mean of dependent variable	1.10	1.10	1.10	1.10	1.10	1.10
Estimation	Poisson	Poisson	Poisson	Poisson	Poisson	Poisson
District & Year FE	✓	✓	✓	✓	✓	✓
Spatial controls		✓		✓		✓

Notes: The outcome is the count of UCDP events located to a district or better, estimated by Poisson on 1990–2025 — the window the routed-water panel spans, less its first year for the lag — on the districts carrying a CaMa river cell, a usable upstream signal and no bifurcation flag. All six columns run on ONE sample, so a coefficient moves between columns because of the controls and never because of the sample. Every column carries the same control set as Table 2: upstream temperature, local rainfall and local temperature, current and lagged. Columns 2, 4 and 6 add the spatial block — neighbours' lagged conflict and neighbours' current and lagged local rainfall and temperature, averaged under row-normalised queen contiguity — so each pair differs in that block alone. Flood-stage days count the days on which the centred three-day mean of upstream-origin discharge exceeded the cell's fitted two-year return level in June–September. The implied-effect rows residualise each treatment on that column's own controls and fixed effects, so the two arms are read in comparable units; a dash marks a column the arm does not enter. Conley spatial-HAC standard errors (150 km cutoff) in parentheses, which is the paper's headline estimator and the basis of the significance stars; Online Appendix Figure B.4 re-prices the headline coefficient under alternative variance estimators. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

arrived leaves the following year's response essentially untouched: whatever carries it is not in the river.

VI. Mechanism: The Opportunity Cost of Fighting

The evidence in this section supports one account. Upstream rain destroys the kharif crop at the planting margin. The loss then reaches households through the labour market rather than through output prices: the farm wage falls, and households respond by supplying *more* labour at exactly the time the non-farm alternative is contracting. The squeeze is strongest in the districts that depend most on agriculture, and it follows the farming calendar — the response appears in the months after the lost harvest would have been sold, not in the days the water takes to travel downstream. What falls is the return to legal work relative to fighting: the opportunity-cost channel of the income–conflict literature (Miguel et al., 2004; Dal Bó and Dal Bó, 2011). Each paragraph below carries one link of that chain; Tables 5 and 6 report the district and household estimates.

Table 5: The mechanism at the district level: four sources of water, one column per outcome

	Kharif area (1)	Cereal prod. (2)	Real ag. wage (3)	Nominal wage (4)	Origin output (5)
log Upstream Precip. (t)	-0.102	-0.232**	-0.081**	-0.066**	0.269***
	(0.071)	(0.106)	(0.037)	(0.031)	(0.049)
log Upstream Precip. $(t-1)$	0.140**	0.045	-0.127***	-0.092***	0.203***
	(0.061)	(0.103)	(0.041)	(0.034)	(0.053)
log Downstream Precip. (t)	0.217***	0.316***	0.054*	-0.030	-0.011
	(0.056)	(0.093)	(0.032)	(0.024)	(0.037)
log Downstream Precip. $(t-1)$	0.159***	0.193**	0.094***	0.028	0.102***
	(0.049)	(0.088)	(0.032)	(0.027)	(0.039)
Estimation	OLS	OLS	OLS	OLS	OLS
District & Year FE	✓	✓	✓	✓	✓
Observations	8,079	7,715	4,556	4,556	8,649
Districts	341	341	276	276	339

Notes: Every column is one regression carrying all four rainfall measures at once — upstream, local, neighbour and downstream, each contemporaneous and lagged — with upstream and local temperatures and their lags, district and year fixed effects, and standard errors clustered on district, as in Table D.7. Each column is estimated on its outcome's own full sample; Table D.7, columns 5–8, reports the same outcomes restricted to the conflict-estimation districts. Only the upstream and downstream terms are shown; local and neighbour rainfall correlate 0.964 after fixed effects and are controls rather than objects of interest. The pattern the text leans on is the mirror image: where upstream and downstream enter opposite-signed and both significant, no confounder common to the two catchments can be responsible, since the same omitted factor cannot raise an outcome from one direction and lower it from the other. All outcomes are logs. Kharif planted area sums hectares over one crop set used by every mechanism exhibit — the six largest kharif crops by planted area (rice, cotton, bajra, maize, groundnut, pigeonpea), the set of Online Appendix Table D.1. As a log of the sum it blends margins: a crop whose plantings are pushed to zero and a crop whose surviving fields shrink both move it; the appendix table separates the margins crop by crop, and the response is concentrated in the extensive one. Cereal production sums tonnes over the four cereals (rice, maize, bajra, jowar): grain tonnes add across cereals, while cotton and groundnut tonnes do not, which is why the production aggregate is narrower than the planted-area one. The price and quantity indices retired from this table remain in Table D.7. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 6: The household after an upstream shock: income, labour allocation and consumption

	Income			Labour allocation			Welfare
	Farm income	Non-farm earnings	Any remittance	Any business	Business days	Farm days	Consumption
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
log Upstream Precip. (t)	0.305	-4.100***	0.004	-0.025	-8.544	-2.369	-0.650***
	(1.529)	(1.259)	(0.076)	(0.055)	(10.912)	(17.640)	(0.177)
log Upstream Precip. ($t-1$)	0.701	-3.424**	0.075	0.030	9.275	4.859	0.054
	(1.686)	(1.430)	(0.064)	(0.028)	(6.540)	(20.670)	(0.172)
log Downstream Precip. (t)	3.242***	1.024	0.044	-0.011	-3.067	25.730**	0.110
	(1.152)	(0.858)	(0.037)	(0.027)	(6.013)	(12.049)	(0.111)
log Downstream Precip. ($t-1$)	2.040*	-0.167	-0.044	0.011	-0.284	-9.108	-0.026
	(1.097)	(0.827)	(0.044)	(0.021)	(4.612)	(9.433)	(0.122)
Estimation	OLS	OLS	OLS	OLS	OLS	OLS	OLS
District & Wave FE	✓	✓	✓	✓	✓	✓	✓
Observations	48,386	48,386	48,386	186,361	186,361	186,361	48,337
Districts	221	221	221	221	221	221	221
Mean of dependent variable			0.11	0.07	16.57	33.08	

Notes: Every column is one regression carrying all four rainfall measures at once — upstream, local, neighbour and downstream, each contemporaneous and lagged — with upstream and local temperatures and their lags, district and survey-wave fixed effects, and standard errors clustered on district, as in Table D.7. Only the upstream and downstream terms are shown; local and neighbour rainfall correlate 0.964 after fixed effects and are controls rather than objects of interest. The pattern the text leans on is the mirror image: where upstream and downstream enter opposite-signed and both significant, no confounder common to the two catchments can be responsible, since the same omitted factor cannot raise an outcome from one direction and lower it from the other. IHDS households (columns 1–3 and 7) and individuals (columns 4–6), 2005 and 2012 waves; the treatment is the mean over each wave's reference years, lagged to the previous wave's. Income: farm income is own-account agricultural income — crop, agricultural property and animal income — accruing to households that operate land or livestock, cushioned by survivor output and cheaper hired labour as well as noisily measured; non-farm earnings fall as displaced labour crowds in; remittance receipt is the transfer response. Labour allocation: whether the person runs a non-farm business, days supplied to it counting non-participants as zero, and farm days on the same convention — the reallocation is into non-farm self-employment, not out of farm work. Welfare: consumption is the log of household consumption; it dips on impact and recovers, and whether the recovery was costly the survey cannot say. Farm and non-farm income are asinh of rupees, participation and remittance receipt are linear probabilities, days are counts; coefficients are not comparable in size across columns. The price of the labour these households sell is in Table 5, and the weekly labour-market margins in NSS are in Online Appendix Tables D.10 and D.11. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

The crop loss.—Planting is the margin that moves: upstream rain cuts kharif planted area while downstream rain raises it by a similar amount, and cereal production follows (Table 5, columns 1–2). The margin is literal — upstream water lowers the probability that each crop is planted at all while acreage in levels moves little, so the loss is plantings pushed to zero where the crop is marginal, not fields shaved at the edges (Online Appendix Tables D.1–D.6). On the origin side, districts upstream of someone raise their own output with their own monsoon rain (column 5): the water that destroys planting below is the same water that grows the harvest above.

Prices versus wages.—The transmission shows in the price of labour, not the price of grain. The aggregate price and quantity indices are insignificant over the full span the crop data allow (Online Appendix Table D.7); crop by crop, the annual farm-harvest price falls only for the coarse cereals and groundnut (Online Appendix Table D.3). The factor price is what gives: the real agricultural wage falls -0.127*** after an upstream shock and rises downstream, and the nominal wage falls with it (Table 5, columns 3–4), so this is pay rather than a deflator artefact. The adjustment runs over about three years (Online Appendix Table D.9).

Labour supply: the dual squeeze.—A falling wage alongside *rising* employment is the signature that separates a supply shift from a demand shift, and three rounds of the National Sample Survey — half a million working-age individuals each, with a one-week reference period — deliver both margins (Online Appendix Tables D.10 and D.11). After an upstream shock, participation in casual agricultural work and the days and earnings from it all rise while the wage falls: households are supplying more labour to a market whose price is dropping, an added-worker response to lost income — farm income first, and the non-farm earnings that fall with it — not prosperity — the coping margin Kochar (1999) documents in hours for idiosyncratic crop shocks and Rose (2001) in participation for district rainfall, moving the wage here because the shock is shared by the whole district — the equilibrium margin Jayachandran (2006) shows is largest exactly where workers are poor, immobile and credit-constrained. When local rain raises output, the wage rises with it — a demand shift (Fetzer, 2020); the upstream shock moves price and quantity apart. Nor is this reallocation into better work: casual non-farm participation, days and earnings all contract at the same horizon, so the exit option is closing as the entry crowds. That dual loss, rather than either margin alone, is the fall in the opportunity cost of fighting these tables measure. What no household survey can see is the conflict margin itself: recruitment on the order of thousands against a labour force of hundreds of millions leaves no trace, so this evidence prices the return to legal work and cannot count the men who leave it.

The household ledger.—The India Human Development Survey — 48,386 households in 221 districts, 2005 and 2012 — follows the same loss into the household (Table 6). Farm income is negative but imprecise, cushioned by survivor output and cheaper hired labour; non-farm earnings fall as displaced labour crowds in; remittances rise the following year, the transfer response an income loss invites. Individuals reallocate into non-farm self-employment — the probability of running a business rises 0.030 and days supplied rise 9.3 — while farm days do not move: distress diversification, entry into low-return work that lowers average non-farm earnings while raising the number earning them. When water fails locally and for good, off-farm reallocation can hold household income together (Blakeslee et al., 2020); here the same monsoon that destroys the crop also contracts the casual non-farm sector, which is what makes the squeeze bind. Consumption dips on impact and is back to flat a year later; whether the recovery was costly, these data cannot say.

Where.—The account requires the response to sit where agriculture is, and it does. Interacting the shock with a pre-window measure of agricultural dependence, the conflict response rises by about half again per standard deviation of dependence and the wage decline deepens with it, upstream and downstream mirror-signed throughout, robust to holding rival moderators — irrigation, poverty, state — fixed (Table 7).

The moderator is fixed at baseline precisely because kharif area is what the treatment destroys; a contemporaneous measure would condition on the outcome.

Table 7: Is the response concentrated where agriculture is?

	Conflict events		Real ag. wage		Nominal wage	
	(1)	(2)	(3)	(4)	(5)	(6)
log Upstream Precip. (t)	-0.316	-0.563	-0.085**	-0.105***	-0.069**	-0.079**
	(0.346)	(0.420)	(0.037)	(0.038)	(0.031)	(0.036)
× Agric. dependence	0.580***	0.381*	-0.066*	-0.125***	-0.017	-0.052
	(0.170)	(0.202)	(0.034)	(0.047)	(0.024)	(0.033)
log Upstream Precip. ($t-1$)	1.121***	1.022**	-0.133***	-0.150***	-0.094***	-0.103***
	(0.258)	(0.409)	(0.040)	(0.041)	(0.034)	(0.034)
× Agric. dependence	0.602***	0.748***	-0.051	-0.099***	-0.047	-0.073**
	(0.158)	(0.157)	(0.031)	(0.037)	(0.030)	(0.030)
log Downstream Precip. (t)	-1.040***	-1.282***	0.051	0.120***	-0.030	-0.002
	(0.349)	(0.342)	(0.032)	(0.039)	(0.024)	(0.028)
× Agric. dependence	-0.283*	-0.524**	0.045**	0.067**	0.011	0.025
	(0.166)	(0.209)	(0.018)	(0.027)	(0.016)	(0.018)
log Downstream Precip. ($t-1$)	-0.971***	-0.988***	0.091***	0.141***	0.025	0.061**
	(0.279)	(0.341)	(0.032)	(0.036)	(0.026)	(0.028)
× Agric. dependence	-0.394	-0.377	0.037	0.067***	0.028	0.043
	(0.260)	(0.267)	(0.026)	(0.024)	(0.032)	(0.032)
Rival moderators		✓		✓		✓
Observations	6,372	6,336	4,556	4,533	4,556	4,533
Number of Districts	177	176	276	275	276	275
Mean of dependent variable	1.06	1.07				
Estimation	Poisson	Poisson	OLS	OLS	OLS	OLS
District & Year FE	✓	✓	✓	✓	✓	✓

Notes: Every column interacts upstream and downstream rainfall, at both horizons, with agricultural dependence; the columns vary the outcome and whether rival moderators are held fixed. Columns 1, 3 and 5 are Poisson on located UCDP events, 1990–2025, under the specification of Table 2, column 4, and the two district wage columns of Table 5 — the male field-labour wage deflated by the local paddy price, and the same wage in nominal rupees — on that table's right-hand side, by OLS. Columns 2, 4 and 6 add THREE rival moderators at once — 2000 irrigated share, 2001 literacy rate and 2001 log worker density — each interacted with the same four rainfall terms, so agricultural dependence is read holding every rival's moderation fixed. Adding them one at a time is reported in the text. The moderator reverses Economic Census non-agricultural employment per head in 2005, so higher means MORE dependent on agriculture. Every moderator here, the rivals included, is standardised, so an interaction reads per standard deviation, and every one is a fixed pre-window cross-section, so none can respond to any year's shock — unlike planted area, which is what the treatment destroys (-0.007, $p = 0.86$). Their own main effects are absorbed by the district effects. The rival columns lose the districts carrying no census covariate. That is not what moves the estimates: on the SAME restricted sample, without the rivals, the lagged interaction is 0.602*** for conflict and -0.051 for the real wage. Standard errors clustered by district. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Timing.—The two accounts part company on the calendar. Figure D.1 replaces the annual lag with twenty-four monthly horizons: conflict is *lower* through the monsoon and the kharif harvest, turns positive in January (0.57) — after the harvest is in and sold, where an income account puts the response and a flood account does not — and peaks in November of $t+1$ (0.91). No single monthly coefficient is precise; the figure is evidence about shape. Nor is the delay the water still travelling: along-channel distance to the upstream basins is a transit of days at plausible river velocities, not weeks. The household side keeps the same calendar: NSS fieldwork quarters sit at fixed

distances from the monsoon by design, and the labour-supply response is absent among households interviewed during it or over the kharif harvest, concentrating among those interviewed eight months after, in the lean season before the next one (Online Appendix Table D.10). The response builds over months and is still present after a year — the market's schedule, not the river's.

Scope.—The mechanism is measured where agricultural data exist. The wage decline and the loss of planted area are rest-of-India results (-0.086, -0.253) and can be neither confirmed nor rejected in Kashmir and the Northeast, where the underlying series are thin. The conflict response, by contrast, is not detectable in the rest of India on its own (0.22, $p = 0.689$): the effect is carried by Kashmir and the Northeast while the mechanism is measured elsewhere, so the two are drawn from largely different districts, and for the Northeast the income channel is untested. One rival carrier can be dismissed directly: weighting origin-district flooding by Census migration shares produces no effect once origins sharing a basin with the destination are dropped (Online Appendix Table E.5) — the shock does not travel through migrant networks.

VII. Infrastructure and Policy

Two policy questions follow naturally from the mechanism. If conflict responds to water the river delivers, then dams — the infrastructure that regulates rivers — might mediate the effect. If the shock works through lost farm income, then India's employment guarantee — a programme built to replace exactly that income — might absorb it. Neither question is settled; this section reports where the evidence currently stands.

Dams.—What I can document is a pattern. Splitting each district's upstream water into a dam-regulated and a free-flowing component, the free-flowing component carries the conflict response, and the contrast survives permutation and placement checks (Online Appendices E.4 and B.5). What I cannot yet do is give the pattern a causal reading. Dams sit on the largest rivers, so no within-district comparison of regulated against free-flowing water separates what a dam *does* from where a dam *is* — and the hydrological first stage runs the wrong way for a buffering story: rainfall on a regulated catchment arrives as more flood downstream, not less, because of where dams are built (Online Appendix Table E.12). I am still working toward an identification strategy that can answer the question; one promising route is to instrument dam placement with river gradient, following Duflo and Pande (2007), and I leave that design to future work.

Public employment.—For the rain that falls on a district's own fields, the safety net works as designed: Fetzer (2020) shows NREGA roughly halves the local rainfall–conflict relationship, his result replicates here exactly, and it survives replacing his rainfall measure with this paper's and his news-based conflict counts with UCDP

(Online Appendix Table D.15). For the water that arrives from upstream, the same programme does nothing detectable. Identified from the staggered roll-out — comparing districts whose programme has already switched on with districts in the same region and year whose programme has not — the upstream–NREGA interaction is zero, and the upstream elasticity is essentially unchanged before and after (Online Appendix Table D.16). The household data agree: the workers the shock displaces do not move onto the programme, and stripping programme earnings out of the ledger barely changes the measured loss (Online Appendix D.3). A reading consistent with all three findings is administrative — relief is triggered by local rainfall deficit and drought declaration, and an upstream shock destroys a crop while local rainfall looks unremarkable — but confirming it requires administrative programme data I have not yet obtained. For now the finding is the gap itself: the absorber that works for local rain does not reach the shock this paper studies.

VIII. Conclusion

This paper reads one monsoon at two places. Rain that falls on a district's own fields lowers violence there; rain that falls on the catchments draining into it — rain the district never saw — raises violence the following year, with an elasticity of 0.91. The two channels behind that contrast now have evidence attached. The river delivers a hazard within the year: discharge at flood stage, plantings prevented at the margin. The market delivers an income loss a season later, and it travels through the labour market rather than through prices: the farm wage falls, households respond by supplying more labour just as the non-farm alternative contracts, and the return to legal work drops on both margins at once. The calendar separates the channels — violence is quiet while the water stands, and turns upward in the months after the lost harvest would have been sold.

The result carries its qualifications with it. It concerns the intensity of ongoing conflict, in one country, over three and a half decades; in the Maoist belt it appears only as a shift from battles toward violence against civilians. The design has no clean outcome-side falsification — the interstate dyad is too thin to carry a verdict — so the reduced form rests on the randomization test over the district–catchment map (Online Appendix B.5) and the fixed-effect variants. The downstream twin of the treatment enters negatively rather than at zero; the two catchment measures are collinear readings of one river corridor, so this is as consistent with corridor spillovers the design absorbs as with a failed placebo, and it is the design's weakest link. The Northeast, where the response is largest, is also where the agricultural series are thinnest and the income channel untested. Online Appendix E collects the diagnostics.

Two implications follow for how climate–conflict relationships are measured. First,

local rainfall is the wrong sufficient statistic wherever drainage networks are large: the local and transmitted components of one monsoon enter with opposite signs, so a design using local rain alone estimates a blend of channels that partly cancel. Second, the transmission runs on two networks — the river within the year, the labour market a season later — and neither is visible to a design that measures rainfall where the violence occurs.

The open questions are the agenda. Whether dams dampen or redirect the water's effect needs placement identified, and instrumenting it with river gradient (Duflo and Pande, 2007) is the design I am pursuing. Whether the employment guarantee could reach upstream-origin shocks is answerable with administrative programme data. Joining the mechanism to the reduced form in the Northeast, and extending the transmitted-shock logic to conflict onset and to other basins, are the natural next steps. The policy stake is already visible in the pattern: relief in India is triggered by the rain that falls, and this paper's shock arrives in districts where no unusual rain fell — the safety net is aimed at the wrong map.

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